DESIGN AND IMPLEMENTATION OF A SEVEN SEGMENT DISPLAY SYSTEM USING 8051

A MINI PROJECT REPORT

Submitted by

DHARSHINI.S **312325106040**

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING





St. JOSEPH'S COLLEGE OF ENGINEERING

(An AUTONOMOUS INSTITUTION)

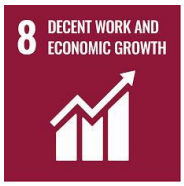
CHENNAI – 600 119

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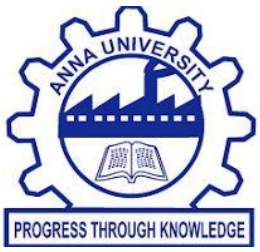
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MINI PROJECT EVALUATION SHEET

Project Title	**:**	
Student Name	**:**	
Register Number	**:**	
Date of Evaluation	**:**	

Evaluation Criteria

S.No	Criteria	Maximum Marks	Marks Awarded
1	**Project Concept & Relevance**	10	
2	**Design & Implementation**	10	
3	**Presentation & Documentation**	10	
Total		30	
Total			

Evaluator Name and Signature

ABSTRACT

This project presents the design and implementation of a seven segment display system using the 8051 microcontroller, developed to provide simple, reliable, and programmable numeric display. The proposed system interfaces a seven segment display unit with the 8051 through its parallel I/O ports. The microcontroller stores the segment code of each digit in program memory and sends the required pattern to the display, so that decimal digits from 0 to 9 can be shown in a controlled sequence. Software timing delays are used to set the counting speed, and the design is extended to a multiplexed multi-digit display in which the digits are refreshed rapidly so that all of them appear continuously lit.

The 8051 serves as the central control unit, managing look-up table access, port output, digit selection, and timing. Current-limiting resistors protect the display segments and the port pins. The proposed system avoids dedicated decoder hardware, giving a compact, low-cost, and easily programmable platform. The project demonstrates the practical application of embedded systems, port interfacing, look-up table programming, timer-based delay generation, and display multiplexing, making it suitable for counters, digital clocks, timers, instrument read-outs, and educational embedded-system demonstrations.

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LIST OF SYMBOLS

SYMBOL	MEANING
V	VOLTAGE
I	CURRENT
R	RESISTANCE
P	POWER
T	TIME PERIOD
D	DUTY CYCLE
VCC	SUPPLY VOLTAGE
GND	GROUND
f	REFRESH FREQUENCY

LIST OF ABBREVIATIONS

ABBREVIATIONS	MEANING
8051	8-BIT MICROCONTROLLER (MCS-51 FAMILY)
LED	LIGHT EMITTING DIODE
SSD	SEVEN SEGMENT DISPLAY
I/O	INPUT / OUTPUT
ROM	READ ONLY MEMORY
RAM	RANDOM ACCESS MEMORY
CC / CA	COMMON CATHODE / COMMON ANODE
BCD	BINARY CODED DECIMAL
MUX	MULTIPLEXING
DPTR	DATA POINTER

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The increasing demand for compact, low-cost, and reliable electronic systems has made numeric displays an essential part of everyday equipment such as digital clocks, counters, weighing scales, meters, and appliance control panels. Conventional display arrangements often rely on fixed hardware, dedicated decoder chips, or manually operated circuits, which offer limited flexibility and increase component count and wiring complexity. These limitations create an opportunity for microcontroller-based systems that can drive displays directly through programmable logic.

The 8051 microcontroller remains one of the most widely taught and used 8-bit microcontrollers. It provides four 8-bit parallel ports, on-chip timers, internal RAM, and program memory, which together make it well suited to interfacing displays. A seven segment display, consisting of seven LED segments (a to g) and an optional decimal point, can show decimal digits and a limited set of characters, and is one of the simplest ways to present numeric output.

This project is motivated by the need for a simple, low-cost, and flexible display system that demonstrates the practical integration of embedded programming and hardware interfacing. The proposed system uses an 8051 to drive a seven segment display, showing digits 0 to 9 in sequence and extending to a multiplexed multi-digit display.

The key idea is to replace dedicated decoder hardware with a software look-up table stored in the 8051's program memory. The same hardware can then show different digits, counting patterns, and speeds simply by changing the program. The project demonstrates port interfacing, look-up table programming, timer-based delay generation, and display

multiplexing, giving a scalable foundation for counters, clocks, timers, and other embedded display applications.

Seven segment displays are used in a very wide range of products, including digital clocks, microwave ovens, washing machines, lift floor indicators, petrol pump meters, weighing scales, vending machines, and laboratory instruments. They are popular because they are bright, easy to read from a distance, inexpensive, and need very little driving circuitry. When such a display is driven by a microcontroller, the same hardware can be reused for many products simply by loading a different program, which shortens the design time and reduces the cost of the final system.

For students of electronics and communication engineering, the seven segment display is an ideal first interfacing exercise. It introduces the use of parallel ports, binary-to-pattern conversion, delay generation, and the idea of persistence of vision in one compact experiment. The project therefore also serves as a useful teaching aid for understanding how an embedded controller interacts with the physical world.

1.2 TECHNOLOGICAL APPROACH

The proposed system adopts a software-driven display approach designed for stable and flicker-free operation. At its core, an 8051 microcontroller acts as the central processing and control unit. One 8-bit port is connected to the segment lines (a to g and dp) of the seven segment display through current-limiting resistors, while additional port pins are used to select the active digit when more than one display is used.

The segment pattern for each digit is stored as a byte in a look-up table in program memory. To show a digit, the 8051 reads the corresponding byte and writes it to the segment port. Software delays, generated with timer or loop routines, control how long each digit remains visible, which sets the counting speed.

For multi-digit operation, the multiplexing technique is used. All displays share the same segment lines, and only one digit is enabled at a time. The 8051 cycles through the digits so quickly that, owing to persistence of vision, all of them appear to be lit continuously. This reduces the number of required I/O pins considerably.

The circuit is built on a breadboard to keep the hardware simple and easy to modify. The performance of the system is evaluated based on display correctness, counting accuracy, flicker-free refresh, reliability under continuous operation, and power consumption.

The development of the system follows a simple flow. The program is first written in assembly language and assembled into a hex file using an 8051 assembler or an integrated development environment such as Keil uVision. The circuit can be checked in a simulation tool before it is built. The hex file is then loaded into the on-chip flash memory of the AT89C51 using a suitable programmer, after which the microcontroller runs the program as soon as power is applied.

The hardware is divided into three parts: the controller section with its clock and reset circuit, the segment drive section consisting of the port and resistors, and the display section with the digit-select lines. This division keeps the design easy to test, because each part can be checked separately before the complete system is operated.

1.3 CHALLENGES AND VISION

Several practical challenges arise in developing a seven segment display system: selecting the correct segment codes for the display type (common cathode or common anode), choosing proper current-limiting resistors, avoiding overloading the 8051 port pins, and maintaining stable timing. The 8051 ports have limited current drive capability, so Port 0 (open-drain) needs external pull-up resistors if it is used, whereas Ports 1, 2, and 3 have internal pull-ups. In a multiplexed display, an inadequate refresh rate causes visible flicker, and unequal on-times produce uneven brightness.

Power management is another concern. Lighting several segments simultaneously increases current consumption, so resistor values must be chosen to limit segment current to a safe level (typically 5-10 mA per segment) while giving adequate brightness. Where several digits share a common line, the total current through the digit-select pin must be considered, and a transistor driver may be required.

The long-term vision of the project is to evolve it into a more capable display platform. The same architecture can be extended to more digits, BCD-to-seven-segment decoders, shift-register based drivers, keypad input, real-time clock integration, temperature and sensor read-outs, and serial-port control from a computer. Such enhancements would make the design suitable for digital clocks, stopwatches, industrial counters, and measurement instruments.

A further challenge concerns the current capability of the 8051 ports. The pins of the 8051 family can sink considerably more current than they can source, because a logic high is produced through a weak internal pull-up. For this reason, many practical designs use a common anode display, so that the port pin sinks the segment current, or place a buffer or driver IC such as a 74HC245 or ULN2003 between the port and the display. The prototype in this project keeps the segment current modest so that the display remains clearly visible while the port pins stay within safe limits.

Software timing is also a challenge. A simple delay loop keeps the processor busy and prevents it from doing other work, while a timer-based delay frees the processor but needs careful calculation of the reload value. Choosing between these two approaches, and selecting a refresh period that is short enough to avoid flicker, are important design decisions that are discussed in the later chapters.

1.4 OBJECTIVES

The primary objectives of this project are:

- To design and implement a seven segment display system using the 8051 microcontroller.
- To develop a look-up table based program that displays the digits 0 to 9 on the display.
- To generate software delays using the 8051 timers so that the counting speed can be controlled.
- To implement a multiplexed multi-digit display with flicker-free refresh using a single shared segment port.
- To evaluate the system based on display accuracy, refresh stability, reliability, and power consumption.
- To demonstrate the practical application of embedded systems, port interfacing, and microcontroller-based display control.

These objectives aim to develop a compact, low-cost, reliable, and scalable display system suitable for counters, clocks, instruments, and educational use.

1.5 ORGANISATION OF THE REPORT

This report is organised into five chapters that present the design, development, implementation, and evaluation of the proposed seven segment display system using the 8051.

Chapter 1 – Introduction presents the background, motivation, technological approach, challenges, vision, and objectives of the project.

Chapter 2 – Literature Survey reviews microcontroller-based embedded systems, the 8051 architecture, seven segment interfacing, multiplexing, and existing display and counter systems.

Chapter 3 – Methodology describes the workflow, system architecture, hardware components, port interfacing, look-up table, timing, and multiplexing software.

Chapter 4 – Results and Discussion presents the experimental setup and the evaluation of display output, multiplexing, timing, reliability, and power consumption.

Chapter 5 – Conclusion and Future Scope summarises the findings and discusses possible improvements such as more digits, decoder and driver ICs, real-time clock integration, and serial or wireless control.

1.6 CONCLUSION

This chapter introduced the background and motivation for developing a seven segment display system using the 8051 microcontroller. The technological approach based on port interfacing, a look-up table, timer delays, and multiplexing was outlined along with the main challenges. The objectives were defined to guide the development and evaluation of the system. The next chapter presents the literature survey.

CHAPTER 2
LITERATURE SURVEY

2.1 EVOLUTION OF MICROCONTROLLER BASED EMBEDDED SYSTEMS

Microcontroller-based embedded systems have become an essential part of modern electronics because they provide dedicated control, automation, and real-time processing in compact hardware. Early digital systems used discrete logic gates, counters, and decoder ICs to perform fixed tasks; any change in function required a change in hardware. The introduction of microcontrollers integrated a processor, memory, timers, and I/O ports on a single chip, so that functionality could be changed through software.

The 8051, introduced by Intel in 1980, became a landmark in this evolution. Its simple architecture, bit-addressable memory, and rich instruction set made it popular in education and industry, and many manufacturers continue to produce compatible devices. Later microcontrollers added more memory, faster cores, and integrated communication features, but the 8051 remains a foundation for learning embedded concepts.

In display applications, microcontrollers replaced dedicated counter and decoder circuits by generating display patterns directly in software. This reduces hardware, lowers cost, and gives flexibility for adding new features. The evolution of microcontroller technology therefore provides a strong foundation for the proposed seven segment display system.

The MCS-51 family has grown into a large group of devices from many manufacturers. The original 8051 had 4 KB of mask-programmable ROM, and the 8031 had no internal program memory at all. Later versions such as the 8052 increased the memory and added a third timer, while flash-based devices such as the AT89C51 and AT89S52 made it easy

to reprogram the chip many times, which is very convenient for students and prototype developers. This long history of compatible devices means that programs and knowledge gained on the 8051 can still be applied widely.

2.2 8051 ARCHITECTURE AND FEATURES

The 8051 is an 8-bit microcontroller from the MCS-51 family. A typical device such as the AT89C51 provides an 8-bit CPU, 4 KB of on-chip program memory, 128 bytes of internal RAM, four 8-bit parallel I/O ports (P0-P3), two 16-bit timers/counters, a full-duplex serial port, and five interrupt sources. It is commonly operated with an 11.0592 MHz crystal, which also allows standard serial baud rates.

The parallel ports allow the microcontroller to interface directly with devices such as LEDs, switches, and displays. Ports 1, 2, and 3 contain internal pull-up resistors, while Port 0 is open-drain and requires external pull-ups when used as an output port. Each pin can source or sink only a limited current, so current-limiting resistors or driver transistors are needed for loads such as LED segments.

The on-chip timers (Timer 0 and Timer 1) can be programmed in several modes to generate accurate time delays without wasting CPU cycles in software loops. With an 11.0592 MHz crystal, one machine cycle is 12 / 11.0592 MHz = 1.085 μs, which is used to calculate timer reload values.

The combination of parallel ports, timers, and a simple instruction set makes the 8051 an appropriate controller for the proposed system. A single device can drive the segment lines, select digits, and generate refresh timing, reducing hardware complexity and allowing future expansion.

The internal RAM of the 8051 is organised into register banks, a bit-addressable area, and a general-purpose scratchpad area. The first 32 bytes hold four register banks of eight registers each (R0 to R7), the next 16 bytes can be accessed bit by bit, and the remaining

bytes are available for data and the stack. Above the RAM lies the group of special function registers (SFRs), which include the port latches P0 to P3, the timer registers TMOD, TCON, TH0, TL0, TH1 and TL1, the accumulator, and the program status word.

The program memory is accessed through the MOVC instruction, which allows constant tables stored in ROM to be read by the program. This feature is used in the proposed system to hold the table of segment codes. Because the table is stored in program memory, it uses no RAM and cannot be altered accidentally while the program is running.

2.3 SEVEN SEGMENT DISPLAY INTERFACING AND CONTROL TECHNIQUES

A seven segment display consists of seven LED segments, labelled a to g, arranged in the shape of the digit 8, plus an optional decimal point (dp). Lighting different combinations of segments forms the digits 0 to 9 and some letters. Displays are available in two types. In a **common cathode** display, all cathodes are joined to ground and a segment lights when its anode is driven high. In a **common anode** display, all anodes are joined to VCC and a segment lights when its cathode is driven low.

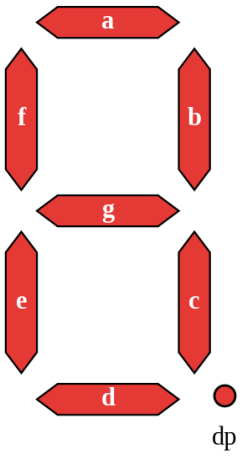


Fig 2.1 Seven Segment Display Segment Labelling

Each segment behaves as an LED and must be connected through a current-limiting resistor. For a 5 V supply, a forward voltage of about 2 V, and a desired current of about 10 mA, the resistor value is R = (5 − 2) / 10 mA = 300 Ω; a standard value of 330 Ω is commonly used.

Several control techniques exist. A BCD-to-seven-segment decoder such as the 7447 or 7448 converts a 4-bit BCD input to segment signals in hardware. Alternatively, the microcontroller can output the segment pattern directly using a look-up table, which saves the decoder IC and allows custom characters. For several digits, static drive (one port per digit) uses many pins, whereas multiplexing shares the segment lines and uses far fewer pins.

The proposed system uses direct microcontroller drive with a look-up table, and multiplexing for multi-digit operation.

Other display technologies are also used with microcontrollers. A 16 × 2 character LCD can show letters and numbers and needs only a few control lines, but it requires an initialisation sequence and is less readable at a distance. Dot-matrix and OLED displays give much greater flexibility but need more memory and more complex software. A seven segment display, in contrast, needs no controller chip of its own, is very bright, and is perfectly suited to purely numeric output, which is why it remains the first choice for counters, timers, and meters.

2.4 DISPLAY MULTIPLEXING AND TIMING

Multiplexing is a technique in which several displays share the same segment lines, and only one display is enabled at any instant. The microcontroller outputs the segment code for one digit, enables that digit for a short interval, disables it, and moves to the next digit. When the whole cycle repeats fast enough, persistence of vision makes all digits appear to be lit at once.

For a flicker-free display, each digit should be refreshed at least about 50 to 60 times per second. For a four-digit display, the digit-switching frequency is therefore at least 4 × 60 = 240 Hz, which corresponds to roughly 4 ms per digit or less. Because each digit is lit for only a fraction of the cycle (duty cycle D = 1/N for N digits), the segment current is usually set somewhat higher than in static operation to preserve brightness, while remaining within the LED and port limits.

Accurate timing is therefore important. On the 8051, either software delay loops or Timer 0 / Timer 1 in the 16-bit mode (Mode 1) can be used to produce the required interval, with the timer approach being more precise and leaving the CPU free for other tasks.

2.5 EXISTING DISPLAY AND COUNTER SYSTEMS

Numeric display and counter systems have progressed from mechanical counters and discrete-logic circuits to programmable electronic systems. Conventional digital counters use counter ICs such as the 7490 or 4026 together with decoder drivers to drive the display. These are simple but fixed in function; changing the count range or format requires hardware changes.

Microcontroller-based systems offer far more flexibility. Common implementations use a microcontroller to display counts, clock time, or sensor values on one or more seven segment displays. However, many basic projects only demonstrate a single static digit or a simple 0-9 count and do not show multiplexed multi-digit operation with careful timing.

The proposed system addresses this gap by combining a look-up table display of digits 0-9 with a multiplexed multi-digit display using only a single 8-bit segment port, forming a simple and scalable platform that can later be extended to clocks, timers, and sensor read-outs.

Several approaches have been reported for driving multi-digit displays from microcontrollers. One approach uses shift registers such as the 74HC595, which convert

serial data from the microcontroller into parallel segment data and so reduce the number of pins needed. Another uses dedicated display driver ICs such as the MAX7219, which handle multiplexing and brightness internally. These solutions simplify the software but add cost and components. Direct multiplexing by the microcontroller, as used in this project, needs no extra driver IC for small displays and gives a clear understanding of the underlying timing, which is valuable in an educational context.

CHAPTER 3
METHODOLOGY

3.1 LIMITATIONS OF EXISTING DISPLAY SYSTEMS

Conventional numeric display systems built from discrete logic have several limitations. Their function is fixed by hardware, so changing the counting sequence, speed, or display format requires rewiring or replacing ICs. Every digit typically needs its own counter and decoder, which increases component count, board area, cost, and power consumption.

Static driving of multi-digit displays needs eight I/O lines per digit, which quickly exhausts the available pins of a microcontroller. Wiring becomes complex and error-prone as digits are added. Basic microcontroller examples may also use blocking delay loops that prevent the processor from doing other work, and may give visible flicker when the refresh timing is not controlled.

These limitations create a need for a compact, programmable solution that controls several digits with few I/O lines and flexible timing. The proposed 8051-based system addresses them through a software look-up table, timer-based delays, and multiplexing.

3.2 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed system replaces dedicated decoder hardware with a software look-up table, making the display fully programmable. New characters or counting sequences can be added by modifying the program without changing the hardware.

Multiplexing allows many digits to be driven using one shared segment port and a few digit-select lines, which greatly reduces pin usage and wiring. The 8051 timers provide accurate and repeatable timing, giving stable, flicker-free refresh.

The hardware is low-cost, compact, and easy to build on a breadboard, with a minimal component count (microcontroller, crystal, reset circuit, resistors, and display). The design is scalable: more digits, driver transistors, keypads, sensors, or a real-time clock can be added with little change to the basic architecture. It is also well suited to educational demonstrations of embedded programming.

Overall, the system provides a cost-effective, flexible, and scalable solution by combining microcontroller control, look-up table programming, and display multiplexing in a single platform.

3.3 FLOWCHART

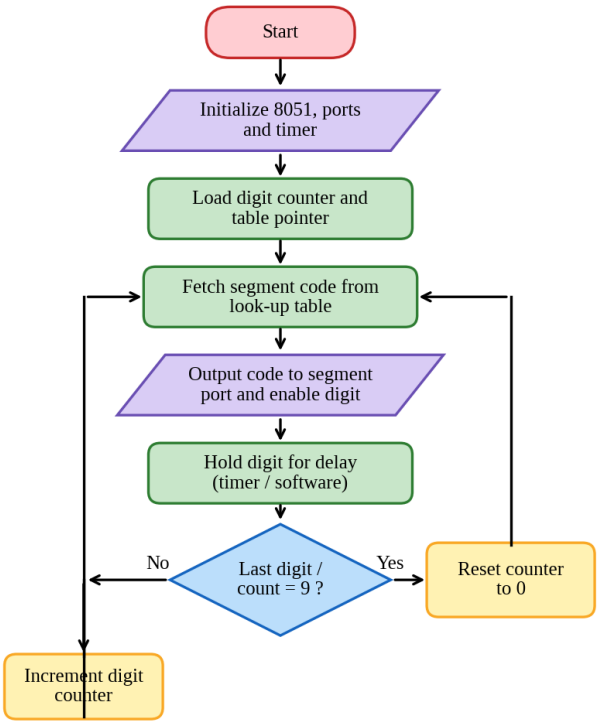


Fig 3.1 Workflow Diagram

The workflow diagram (Fig. 3.1) illustrates the end-to-end operation of the proposed system. The process begins with initialization of the 8051: the stack pointer is set, the

segment port and digit-select lines are configured, and the timer is initialized. A pointer to the look-up table and a digit counter are then loaded.

For the current digit value, the 8051 fetches the corresponding segment code from the look-up table in program memory and writes it to the segment port. In single-digit mode, the code is held for the programmed delay interval, after which the counter is incremented. When the count passes 9, it rolls back to 0 and the sequence repeats.

In multiplexed mode, the 8051 outputs the segment code of one digit, enables only that digit's select line for a short interval, then disables it and proceeds to the next digit. After the last digit, the cycle restarts. This continuous loop keeps all digits refreshed and gives a stable display.

The flow of the program is a continuous loop, so the display never stops once power is applied. A manual reset or a power failure returns the program to the initialization block, after which the sequence starts again from digit 0. Because the look-up table is read for every digit, the same loop structure serves both the single-digit counting program and the multiplexed multi-digit program, with only the number of digits and the delay value being different.

3.4 SYSTEM ARCHITECTURE

The system follows a modular structure consisting of the clock and reset unit, the 8051 processing unit, the look-up table, the segment driver (port and resistors), the digit-select section, and the seven segment display output.

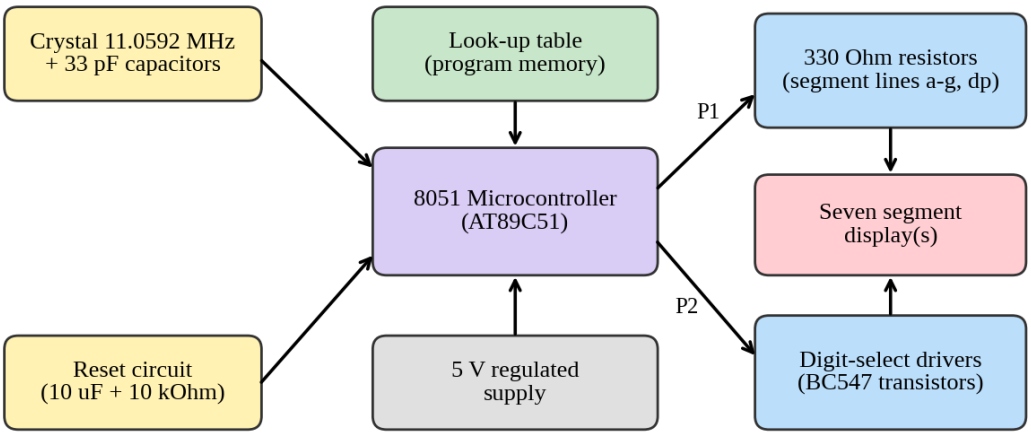


Fig 3.2 8051 Based System Architecture

The 8051 acts as the central control unit. It reads the segment code for each digit from the look-up table in program memory and sends it through the segment port, via current-limiting resistors, to the display segments. In multiplexed operation it also drives the digit-select lines, which enable one display at a time, directly for low-current loads or through transistors for higher current.

A crystal oscillator with two 33 pF capacitors provides the clock, and a power-on reset circuit (10 µF capacitor with a 10 kΩ resistor) ensures correct start-up. The external-access pin (EA) is tied to VCC so the microcontroller executes from on-chip program memory.

This modular architecture simplifies implementation, testing, and debugging, and allows additions such as more digits, decoder drivers, keypads, sensors, or serial communication with minimal change.

The power supply section provides a regulated 5 V to the 8051 and the display. A decoupling capacitor close to the supply pins of the microcontroller helps to suppress noise caused by the switching of the display currents. The segment lines are driven from the Port 1 latch, while the digit-select lines are driven from Port 2. Because the segment lines are shared by all digits, the wiring between the microcontroller and the display is the same whether one digit or four digits are connected.

3.5 HARDWARE COMPONENTS AND SPECIFICATIONS

The system uses an 8051-family microcontroller (AT89C51) as the central controller. The seven segment display is connected to Port 1 through current-limiting resistors. A crystal oscillator, capacitors, and a reset circuit provide clock and start-up. The circuit is assembled on a breadboard using jumper wires and is powered from a regulated 5 V supply.

Table 3.1 Hardware Components and Specification

COMPONENT	SPECIFICATION / DESCRIPTION
8051 MICROCONTROLLER (AT89C51)	8-bit microcontroller, 4 KB flash, four I/O ports, two timers
SEVEN SEGMENT DISPLAY	Common cathode, 7 segments + decimal point (single and 4-digit units)
RESISTORS (8 × 330 Ω)	Current-limiting resistors for display segments
RESISTOR (10 kΩ)	Reset circuit pull-down
CRYSTAL OSCILLATOR	11.0592 MHz
CAPACITORS (2 × 33 pF)	Crystal oscillator load capacitors
CAPACITOR (10 µF)	Power-on reset
PUSH BUTTON	Manual reset
NPN TRANSISTORS (BC547)	Digit-select drivers for multiplexed operation
BREADBOARD AND JUMPER WIRES	Solderless prototyping and connections
5 V REGULATED SUPPLY	Power source for the system

The selected components provide a low-cost, flexible platform for implementing the display system, and the modular arrangement allows later expansion.

The 8051 is the heart of the design and executes the program stored in its flash memory. The crystal and its two capacitors set the clock frequency, and the 11.0592 MHz value is chosen because it divides exactly into the standard serial baud rates, which keeps the

design compatible with later serial-communication extensions. The 10 µF capacitor and 10 kΩ resistor form the power-on reset circuit that holds the reset pin high for a short time after power is applied. The 330 Ω resistors limit the current of each segment, and the BC547 transistors act as switches that connect the common terminal of the selected digit to ground in the multiplexed version.

3.6 8051 BASED SEVEN SEGMENT INTERFACING

The segment lines a to g and dp of the display are connected to port pins P1.0 to P1.7 through 330 Ω resistors. The digit-select lines for the multiplexed display are driven from lower pins of Port 2. The pin assignment is given below.

Table 3.2 8051 Port Pin Configuration for Seven Segment Display

DISPLAY PIN	8051 PIN	FUNCTION
a	P1.0	Segment a
b	P1.1	Segment b
c	P1.2	Segment c
d	P1.3	Segment d
e	P1.4	Segment e
f	P1.5	Segment f
g	P1.6	Segment g
dp	P1.7	Decimal point
Digit 1-4 common	P2.0 – P2.3	Digit select (through transistors)

For a common cathode display, a segment glows when its port pin is at logic 1. The segment codes (with P1.7 = dp = 0) are:

Table 3.3 Segment Codes for Digits 0-9

DIGIT	PATTERN (dp g f e d c b a)	CODE (COMMON CATHODE)	CODE (COMMON ANODE)
0	0011 1111	3FH	C0H

DIGIT	PATTERN (dp g f e d c b a)	CODE (COMMON CATHODE)	CODE (COMMON ANODE)
1	0000 0110	06H	F9H
2	0101 1011	5BH	A4H
3	0100 1111	4FH	B0H
4	0110 0110	66H	99H
5	0110 1101	6DH	92H
6	0111 1101	7DH	82H
7	0000 0111	07H	F8H
8	0111 1111	7FH	80H
9	0110 1111	6FH	90H

The 330 Ω resistors limit each segment current to about 9 mA at 5 V, which protects both the display and the port pins. For a common anode display, the same circuit can be used with the common anode codes and the common pin connected to VCC.

The current taken by the display can be estimated from the resistor value. With a 5 V supply, a segment voltage drop of about 2 V, and a 330 Ω resistor, each lit segment draws about 9 mA. When all seven segments of a digit are lit (digit 8), the digit draws about 63 mA, and this total current flows through the common pin of the digit. Since this is larger than a port pin can safely handle, the common pin of each digit is switched through a transistor in the multiplexed version, which also keeps the digit-select lines of the 8051 lightly loaded.

3.7 SOFTWARE DESIGN AND MULTIPLEXING

The program is written in 8051 assembly language (it can equally be written in embedded C) and stored in the on-chip flash memory. The segment codes are placed in a look-up table using the DB directive, and the data pointer (DPTR) together with the MOVC

instruction fetches the code for each digit. The single-digit counting program for a common cathode display is given below.

```
        ORG  0000H
MAIN:   MOV  R0, #00H          ; digit counter
LOOP:   MOV  DPTR, #TABLE      ; point to look-up table
        MOV  A, R0
        MOVC A, @A+DPTR        ; fetch segment code
        MOV  P1, A             ; display digit
        ACALL DELAY            ; hold for set time
        INC  R0
        CJNE R0, #0AH, LOOP    ; repeat until 10
        SJMP MAIN              ; restart from 0
DELAY:  MOV  R1, #05H
D1:     MOV  R2, #0FFH
D2:     MOV  R3, #0FFH
D3:     DJNZ R3, D3
        DJNZ R2, D2
        DJNZ R1, D1
        RET
TABLE:  DB 3FH,06H,5BH,4FH,66H,6DH,7DH,07H,7FH,6FH
        END
```

In the multiplexed mode, the main routine repeats the following steps for each digit: place the segment code for that digit on P1, enable the corresponding digit-select line on P2, wait about 2 to 4 ms using Timer 0 in Mode 1, then disable the digit before moving to the next. With four digits and about 2.5 ms per digit, one complete cycle takes 10 ms, giving a refresh rate of about 100 Hz per digit, well above the flicker threshold. Turning the digit off before changing the segment data prevents ghosting between neighbouring digits.

The counting speed in single-digit mode is changed simply by modifying the delay constants, and the displayed value in multiplexed mode is obtained by splitting the number into digits and fetching each code from the same table.

The timer delay used for multiplexing is calculated from the machine cycle. With an 11.0592 MHz crystal, one machine cycle is 1.085 µs. For a digit on-time of about 2.5 ms, the timer must count 2500 / 1.085, which is about 2304 cycles. In Mode 1 the timer counts up from the loaded value to 65536, so the reload value is 65536 − 2304 = 63232, which is F700H. Therefore TH0 is loaded with F7H and TL0 with 00H. The complete four-digit multiplexed program is given below.

```
        ORG  0000H
```

```
START:  MOV  TMOD, #01H        ; Timer 0, Mode 1 (16-bit)
        MOV  30H, #01H         ; digit 1 value
        MOV  31H, #02H         ; digit 2 value
        MOV  32H, #03H         ; digit 3 value
        MOV  33H, #04H         ; digit 4 value (displays 1234)
MAIN:   MOV  R0, #30H          ; pointer to digit buffer
        MOV  R1, #01H          ; select mask, starts at P2.0
NEXT:   MOV  A, @R0            ; read digit value
        MOV  DPTR, #TABLE
        MOVC A, @A+DPTR        ; fetch segment code
        MOV  P1, A             ; put code on segment lines
        MOV  P2, R1            ; enable current digit
        ACALL T2MS             ; keep it on for about 2.5 ms
        MOV  P2, #00H          ; disable digit to avoid ghosting
        INC  R0                ; next digit in buffer
        MOV  A, R1
        RL   A                 ; shift select mask
        MOV  R1, A
        CJNE R1, #10H, NEXT    ; mask becomes 10H after 4 digits
        SJMP MAIN              ; repeat refresh cycle
T2MS:   MOV  TH0, #0F7H
        MOV  TL0, #00H
        SETB TR0               ; start Timer 0
WAIT:   JNB  TF0, WAIT         ; wait for overflow
        CLR  TR0
        CLR  TF0
        RET
TABLE:  DB 3FH,06H,5BH,4FH,66H,6DH,7DH,07H,7FH,6FH
        END
```

In this program the digit values are held in the RAM locations 30H to 33H, so the displayed number can be changed simply by writing new values into these locations.

3.8 DISPLAY OPERATION AND PERFORMANCE EVALUATION

The proposed system displays the digits 0 to 9 in sequence on a single seven segment display and shows multi-digit numbers on a multiplexed four-digit display. The pattern is generated by sending the look-up table codes to the segment port, and different outputs are obtained by changing the program without changing the hardware.

In single-digit operation, each digit is held for a software-defined interval. In multiplexed operation, the digits are refreshed in rapid succession. Timing is controlled through software loops or the 8051 timers.

Performance evaluation verifies correctness and reliability. The main parameters are accuracy of the displayed digits compared with the programmed sequence, counting interval, absence of visible flicker, uniformity of brightness across digits, stability during continuous operation, and power consumption. Display accuracy is checked by visual

comparison of the output with the programmed values, and refresh stability is judged by observing the multiplexed display for flicker and ghosting.

The system is also run continuously to check its reliability. Current-limiting resistors keep segment currents safe and prevent overloading of the port pins.

The test procedure is carried out in stages. First, the circuit connections are checked against the pin table, and the supply voltage at the microcontroller is measured. Next, the single-digit program is loaded, and each digit from 0 to 9 is checked by comparing the glowing segments with the expected pattern. Then the multiplexed program is loaded, and the display is observed for flicker, ghosting, and brightness differences between digits. A result is accepted only when every digit is shown correctly, the sequence repeats without error, and no flicker is visible.

3.9 CONCLUSION

This chapter described the methodology used to implement the seven segment display system using the 8051 microcontroller. The architecture integrates the 8051, a look-up table, segment drive circuitry, and multiplexed digit selection. The workflow, hardware components, pin assignments, segment codes, software, and evaluation parameters were explained. The next chapter presents the experimental results and analysis.

CHAPTER 4
RESULTS AND DISCUSSIONS

4.1 EXPERIMENTAL SETUP

The experimental setup was developed to evaluate the performance of the proposed seven segment display system. The hardware consists of an 8051 (AT89C51) microcontroller, an 11.0592 MHz crystal with two 33 pF capacitors, a reset circuit, a common cathode seven segment display (single and four-digit), 330 Ω current-limiting resistors, digit-select transistors, a breadboard, jumper wires, and a regulated 5 V supply.

The program was written in assembly language, assembled into a hex file, and loaded into the on-chip flash memory of the microcontroller. The segment lines were connected to Port 1 and the digit-select lines to Port 2.

During experimentation, the display was first tested with the single-digit counting program, followed by the multiplexed multi-digit program. The system was operated repeatedly to verify the consistency of the displayed digits and the stability of the refresh. The setup was also used to analyse display accuracy, timing, reliability, and power consumption, which are discussed in the following sections.

The program was developed and assembled on a personal computer, and the circuit was first checked in a simulation tool before the hardware was built. After the hex file was loaded into the microcontroller, the supply was applied and the behaviour of the display was observed. The supply voltage at the microcontroller pins was verified to be close to 5 V, and the crystal and reset connections were checked before the display program was tested.

4.2 DISPLAY AND COUNTING RESULTS

The system was successfully tested with a seven segment display to verify correct operation of the programmed digits. The 8051 sent the segment codes from the look-up table to Port 1, and each digit from 0 to 9 appeared correctly on the display. After the digit 9, the counter rolled over to 0 and the sequence repeated continuously.

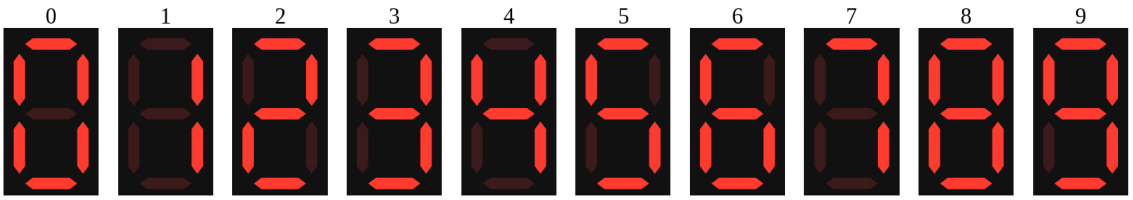


Fig 4.1 Digits 0 to 9 on the Seven Segment Display

The counting interval could be changed by adjusting the delay constants in the program, showing that the speed is controlled entirely by software. No change in the hardware was required to alter the sequence or speed.

Table 4.1 Displayed Digits and Observed Results

DIGIT SENT	CODE (HEX)	SEGMENTS ON	OBSERVED RESULT
0	3FH	a, b, c, d, e, f	Digit 0 displayed correctly
1	06H	b, c	Digit 1 displayed correctly
2	5BH	a, b, d, e, g	Digit 2 displayed correctly
3	4FH	a, b, c, d, g	Digit 3 displayed correctly
4	66H	b, c, f, g	Digit 4 displayed correctly
5	6DH	a, c, d, f, g	Digit 5 displayed correctly
6	7DH	a, c, d, e, f, g	Digit 6 displayed correctly
7	07H	a, b, c	Digit 7 displayed correctly
8	7FH	a, b, c, d, e, f, g	Digit 8 displayed correctly
9	6FH	a, b, c, d, f, g	Digit 9 displayed correctly

Overall, the results confirm that the 8051-based look-up table method produces accurate and stable numeric output on the seven segment display with simple, programmable control.

The counting speed depends on the delay constant R1 in the delay routine. Each pass of the innermost loops takes about 130,800 machine cycles, which is about 0.14 s at 11.0592 MHz, and the loop is repeated R1 times. The hold time of each digit for different delay constants was therefore calculated, and the observed counting speed agreed with these values.

Table 4.2 Delay Constant and Digit Hold Time

DELAY CONSTANT R1	HOLD TIME (CALCULATED)	OBSERVED COUNTING
1	0.14 s	Rapid counting
2	0.28 s	Fast counting
5	0.71 s	Comfortable to read
10	1.42 s	Slow counting

The delay of about 0.7 s used in the program gives a rate at which each digit can be read comfortably, while still showing that the display changes continuously.

Each digit was also checked individually by comparing the glowing segments with the expected pattern from the code table. In every case the segments that lit corresponded exactly to the bits that were set in the code byte, which confirms that the pin connections between Port 1 and the display segments are correct. A wrong connection would have shown up immediately as a distorted digit, for example a 2 appearing like a 5, and no such distortion was observed.

The same program can be used with a common anode display by replacing the table with the complementary codes and connecting the common pin to the supply. This was not

required in the prototype, but it shows that the method is independent of the display type and that only the contents of the table need to change.

4.3 MULTIPLEXED DISPLAY RESULTS

The multiplexed four-digit display was tested to evaluate simultaneous display of several digits using a single shared segment port. The 8051 output the segment code for one digit, enabled that digit for a short interval, and moved to the next digit in a continuous loop.

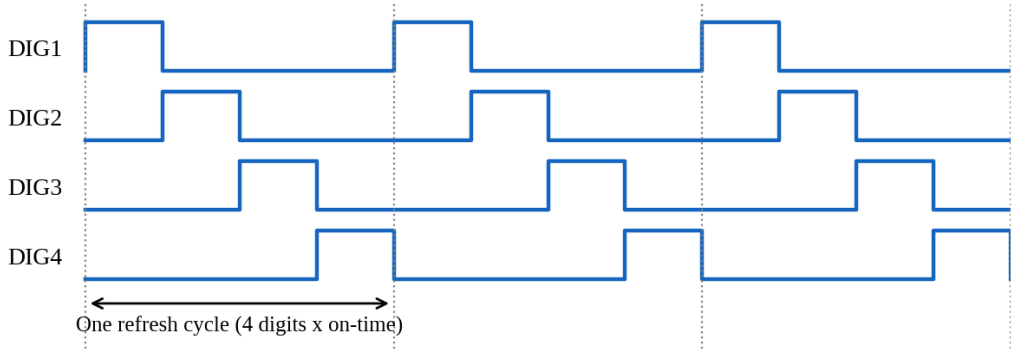


Fig 4.2 Digit Enable Timing in Multiplexed Operation

With a per-digit on-time of a few milliseconds, all four digits appeared to be lit continuously and no visible flicker was observed. Different numbers were displayed by changing the digit values in the program. When the refresh interval was made too long, flicker became noticeable, confirming the need for a sufficiently high refresh rate. Disabling each digit before changing the segment data removed ghosting between adjacent digits.

Table 4.3 Multiplexed Display Observations

CONDITION	OBSERVATION
Per-digit on-time of a few ms	Stable, flicker-free display
Very long per-digit on-time	Visible flicker
Digit disabled before data change	No ghosting
Different numbers loaded	Correct digits shown at correct positions

The results confirm that multiplexing allows several digits to be driven with few I/O lines while maintaining a clear and stable display.

The refresh rate of the multiplexed display depends on the on-time of each digit. For four digits, the refresh rate of each digit is 1 / (4 × on-time). An on-time of 2 ms gives 125 Hz, 2.5 ms gives 100 Hz, 5 ms gives 50 Hz, and 10 ms gives only 25 Hz. The first three values produced a steady display, whereas the 25 Hz value produced clearly visible flicker. The value of 2.5 ms was therefore chosen, as it gives a comfortable margin above the flicker threshold while leaving enough time for the processor to prepare the next digit.

The brightness of a multiplexed digit is lower than that of a statically driven digit, because each digit is lit for only one quarter of the refresh cycle. With four digits the duty cycle is 25 percent, and the average current of each segment is therefore about a quarter of its peak current. Even so, the persistence of vision of the eye makes the display appear steady, and the brightness was found to be adequate for indoor viewing. If higher brightness is needed, the segment current can be increased slightly by using a smaller resistor, provided the peak current remains within the limit of the LED and the driver.

All four digits showed the same brightness, because the same on-time was given to each digit. When the on-time of one digit was deliberately made longer than the others, that digit appeared brighter, which confirms the relation between on-time and perceived brightness.

4.4 TIMING AND SYSTEM RELIABILITY

Timing and reliability were evaluated by running the display continuously and repeatedly changing the displayed values. Timing was considered in terms of the interval each digit remained visible in counting mode and the refresh interval in multiplexed mode. The delays produced by the program were found to be consistent from one cycle to the next, and the timer-based refresh remained steady during long operation.

Table 4.4 Timing and Reliability Observations

PERFORMANCE PARAMETER	OBSERVATION
Digit Accuracy	Displayed digits matched programmed values
Counting Interval	Consistent and adjustable through software
Refresh Stability	No visible flicker in multiplexed mode
Brightness Uniformity	Even brightness across digits
Continuous Operation	Stable during extended testing
Reset Behaviour	Program restarted correctly on power-up and manual reset

The system maintained correct display output without unexpected changes or interruptions. The experimental results indicate that the combination of 8051 processing, look-up table control, and timer-based multiplexing provides reliable and consistent display performance. Further optimisation is possible using interrupt-driven refresh so that the main program can perform other tasks while the display is refreshed in the background.

The system was also switched off and on several times, and the display started from digit 0 correctly each time, which confirms that the reset circuit and the program start-up work properly. The display was left running for an extended period, and no change in the digit pattern, brightness, or timing was noticed. These observations indicate that the design is suitable for continuous use in simple counters and indicators.

Possible sources of error were also considered. The crystal frequency determines every delay in the system, so a slightly different crystal changes the counting speed and the refresh rate by the same proportion. Poor contacts on the breadboard can cause a segment to flicker or fail to light, and a loose connection to the reset circuit can restart the program unexpectedly. These problems were avoided by keeping the wires short, pressing the components firmly into the breadboard, and checking the connections before each test.

4.5 POWER CONSUMPTION AND PERFORMANCE ANALYSIS

The system is powered from a regulated 5 V supply that feeds the microcontroller and the display. Each lit segment draws approximately 9 mA with a 330 Ω resistor, estimated from I = (5 − 2) / 330 = 9 mA. The current drawn by the display therefore depends on how many segments are lit at the time; the digit 8 (all seven segments) draws the most and the digit 1 (two segments) the least. In multiplexed operation, the average current per digit is reduced by the duty cycle, since each digit is on for only a fraction of the refresh cycle.

The 8051 itself draws a small, fairly constant current for its operation. Switching off unused digits and segments in software helps avoid unnecessary power consumption.

Table 4.5 Power Consumption Under Different Operating Conditions

OPERATING CONDITION	DISPLAY ACTIVITY	EXPECTED POWER LEVEL
Reset / initialisation	Display OFF	Low
Digit 1 displayed	2 segments ON	Low
Digit 7 displayed	3 segments ON	Low
Digit 4 displayed	4 segments ON	Low to moderate
Digit 8 displayed	7 segments ON	Moderate
Multiplexed four digits	Digits ON in turn at 1/4 duty	Moderate

The analysis shows that the system provides a good balance between display capability, timing accuracy, and power consumption for a small-scale display application. The 8051 handles port control, timing, and multiplexing within a single compact platform, with no separate decoder IC required. The results also provide a foundation for improvements such as driver ICs for higher current, optimised power management, and larger displays.

Taking about 9 mA per segment, the digit 8 draws about 63 mA, the digit 7 about 27 mA, and the digit 1 about 18 mA. In the multiplexed display only one digit is lit at a time, so

the display current at any instant does not exceed that of a single digit, which is about 63 mA in the worst case. The maximum power drawn by the display is therefore about 5 V × 63 mA = 0.32 W. Together with the small current taken by the microcontroller, the total remains well within the capacity of a small regulated supply.

4.6 CONCLUSION

The implemented system successfully demonstrates a practical and flexible approach to numeric display using the 8051 microcontroller. The system integrates the 8051 with a seven segment display, a look-up table, and timer-based multiplexing, enabling digits 0 to 9 and multi-digit numbers to be shown reliably. The experimental results confirmed correct digit display, adjustable counting speed, and stable, flicker-free multiplexed operation.

Certain limitations were observed. The number of digits that can be driven is limited by the available port pins and the current capability of the 8051, so transistor drivers become necessary for larger displays. The multiplexed display occupies the processor for refreshing unless interrupts are used, and the brightness of a multiplexed display is lower than in static drive because of the duty cycle. Proper resistor selection, driver design, and refresh timing are therefore important.

Overall, the project shows how a low-cost 8051 platform can be used to build a programmable and scalable seven segment display system, forming a base for future enhancements such as clocks, timers, sensor read-outs, and serial-controlled displays.

The results also show that careful choice of the on-time and the delay constants is more important than the speed of the processor, because the display requirements are far slower than the operating speed of the 8051. This leaves a large part of the processing time free, which can be used in future versions for reading keys, sensors, or a serial port while the display is refreshed in the background by a timer interrupt.

The overall observations of this chapter confirm that the objectives set at the beginning of the project were met. The display showed the correct digits, the counting speed could be adjusted, the multiplexed display was free from flicker and ghosting, and the system worked reliably during continuous operation with a low power requirement.

When compared with a decoder-based design, the software look-up table approach was found to need fewer components and to be easier to modify, at the cost of using some processor time and program memory. For a small display this cost is negligible, and the flexibility gained is a clear advantage.

CHAPTER 5
CONCLUSION

5.1 CONCLUSION

This project presented a seven segment display system designed to provide flexible numeric display using the 8051 microcontroller. The system integrates a seven segment display with 8051 port interfacing, a software look-up table, timer-based delays, and display multiplexing.

The developed system was tested with single-digit counting from 0 to 9 and with a multiplexed multi-digit display. The experimental results demonstrated that the 8051 produced correct digits, consistent timing, and stable, flicker-free output. The design achieved this without any dedicated decoder IC, reducing hardware and cost.

Overall, the proposed solution provides a compact, low-cost, and scalable platform for numeric display applications. The integration of microcontroller processing, port interfacing, and timer-based multiplexing demonstrates the usefulness of the 8051 for embedded display systems, and the design can be enhanced with further features for future applications.

5.2 SUGGESTIONS FOR FUTURE WORK

Although the proposed system successfully demonstrated reliable seven segment display control, several improvements can be explored. The number of digits can be increased, using shift registers (such as 74HC595) or dedicated display driver ICs (such as MAX7219) to reduce pin usage and relieve the port current limits.

The display can be extended into a digital clock or stopwatch using a real-time clock IC (such as DS1307), or into a measurement display by adding sensors and an analog-to-

digital converter for temperature, voltage, or light read-outs. A keypad or push buttons can be added for user input, enabling set-points, timers, and menu control.

The 8051 serial port can be used to receive values from a computer or a wireless module such as Bluetooth or Wi-Fi, allowing remote control and IoT integration. Interrupt-driven refresh can free the processor for other tasks, and brightness control with pulse-width modulation or lower-power operating modes can further improve efficiency. Together, these enhancements can turn the prototype into a more advanced and intelligent display system suitable for educational, industrial, and consumer applications.

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